

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1974

Rittenhouse Tap 1 100 kV
October 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1974. The Interconnection Customer proposes a 137.0 MW battery energy storage facility interconnecting at the Rittenhouse Tap 1 100 kV substation in the Illinois area. The request was received on January 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 533 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$527,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

French Landing Dam – Pineville Delivery 1 100 kV #2: 106.9% loading (141 MVA) following the outage of the Pineville Delivery 1 – French Landing Dam 100 kV #1; DFAX = 6.7%.

The Pineville Delivery 1 – French Landing Dam 100 kV L is overloaded to 106.4% of its 142 MVA emergency rating for the loss of the Pineville Delivery 1 Substation – French Landing Dam Substation 100 kV Line 2. The project contributes 6.7% DFAX (9.2 MW) to this overload.

The Battle Mtn – Spills 100 kV Circuit 1 is overloaded to 104.2% of its 176 MVA emergency rating for the loss of the Spills – Battle Mtn 100 kV Cir. 2. The project contributes 8.6% DFAX (11.7 MW) to this overload.

Spills – Battle Mtn 100 kV Cir. 2: 104.2% loading (176 MVA) following the outage of the Spills – Battle Mountain 100 kV; DFAX = 8.6%.

Kimbery – Coldwater 100 kV: 100.0% loading (101 MVA) following the outage of the Tidelands – Fairfield Works 100 kV; DFAX = 7.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
French Landing Dam – Pineville Delivery 1 100 kV #2	Reconductor line	\$399,000	16% (\$64,000)
French Landing Dam – Pineville Delivery 1 100 kV Cir. 1	Reconductor line	\$345,000	16% (\$55,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Spills – Battle Mountain 100 kV Line 1	Reconductor line	\$1,320,000	11% (\$146,000)
Battle Mountain – Spills 100 kV #2	Reconductor line	\$2,050,000	11% (\$227,000)
Kimberly – Coldwater 100 kV	Reconductor line	\$385,000	9% (\$35,000)
	Total allocated to project		\$527,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.